Kéllia Umuhuza Gatete

Mechanical & Aerospace Engineer | Space Systems • Planetary Robotics • Mechatronics • R&D
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EDUCATION

Princeton University, Princeton, NJ **May 2026**
B.S.E., Mechanical & Aerospace Engineering | Minor in Robotics — ABET-accredited programs in Mechanical and Aerospace Engineering.
Relevant Coursework: Space System Design; Space Flight; Autonomous Fabrication & Robotics; Microprocessors for Measurement & Control; Rocket & Air-Breathing Propulsion; Materials Science

TECHNICAL SKILLS

Space & Systems: Ansys STK/Astrogator, spacecraft C&DH architecture, processing/storage/data budgets, orbital and trajectory analysis, link-budget analysis, requirements and traceability, ConOps, trade studies, risk matrices, verification & validation
Robotics & Embedded: Arduino/ATmega328P, Teensy, Raspberry Pi, micro:bit, Hall-effect and temperature sensors, relays, servos, serial communication, assembly programming, PID/feedback control, data acquisition
Design, Test & Fabrication: Fusion 360, Rhino 8, Simplify3D/G-code, CAD/CAM, FEA, 3D printing, manual machining, ABB RobotStudio/RAPID/FlexPendant, MTS testing, thermal/dust/abrasion testing
Programming & Data: Python, MATLAB/Simulink, C++, JavaScript, LabVIEW, SPSS, Overleaf

ENGINEERING & RESEARCH EXPERIENCE

Senior Thesis Researcher — OSCAR MARSKIN, Princeton MAE **Sep 2025 – Apr 2026**
Planetary soft-robotics protection and sensing system; co-authored 130-page thesis

- Co-designed and fabricated MARSKIN, a six-material flexible environmental skin for an origami crawler, combining TPU, EcoFlex, aerogel, PETG-PTFE, aluminium and a superhydrophobic coating; balanced insulation, structural support, adhesion, dust and abrasion protection, and compliant motion while moving from a circular test article to an OSCAR-specific panelized geometry compatible with folding.
- Validated MARSKIN against 2 controls using conductive/radiant heating, 30-min cold exposure, 3 hot–cold cycles, dust and abrasion: across 3 heat-lamp distances, maintained ~24–27 °C peak through-thickness ΔT versus 4.6 °C for controls; at 75 °C conductive heating, improved peak/mean ΔT by ~85% with no visible delamination.
- Built the onboard sensing and data pipeline on a Raspberry Pi Zero 2 W, synchronising contact TMP117 and non-contact MLX90614 measurements at 1 Hz with 1080p day/night imaging every 10 s.

ACEE Summer Research Intern — Ju Lab, Princeton **Summer 2025**
Combustion & Low Carbon Energy Conversion Laboratory | Prof. Yiguang Ju

- Investigated design-stage dielectric-barrier-discharge plasma conversion of CO_2 to graphene at ~400 °C and evaluated 2 optical diagnostics — laser absorption spectroscopy and laser-induced fluorescence — for real-time CO/CO_2 monitoring and reactor optimisation.

Research & Gender Intern — Aegis Trust, Kigali, Rwanda **Summers 2023 & 2024**
Two-year gender-norms programme with the University of Rwanda Centre for Gender Studies | Funded 2× by the Summer Social Impact Internship Fund

- Analysed a 612-participant baseline study in SPSS across 5 districts and triangulated the quantitative results against document review, semi-structured interviews and a reflection workshop; the programme reached 3,753 stakeholders against a 2,600 target.
- Built assessment criteria for programme outcomes and co-developed an awards scheme with the Centre for Gender Studies, selecting 50 recipients through district committees — the deliverables that outlasted the funding cycle.

SELECTED SPACE & ROBOTICS PROJECTS

LUREX Lunar Mission — Operations/Payload & CCD **Spring 2026**
Lunar Radiation Environment Explorer | MAE 342 Space System Design

- **Rebuilt the onboard-computer budget after PDR review**, replacing a percentage-based allocation with a throughput-driven method: sized supervisory workload in KIPS by software function across 5 operating modes, converted to required clock speed, and closed the governing case (relay operations) at 14.29 MHz against 1,000 MHz available — every figure traceable to a measured data rate.
- Sized memory, storage and interfaces against the same rates: 63.5 MiB RAM of 512 available; recorder requirements of 1,040.7 MB and 332.6 MB from a 71.3-min worst-case blackout, verified against a 32 GB recorder at >30× margin; confirmed SpaceWire and CAN throughput under peak concurrent recorder, payload and framing traffic.
- Owned the mission risk matrix and cost compliance analysis across 8 subsystems, reconciling estimates against a $175M SMEX cap plus $5M launch services; moved near-real-time downlink and onboard data storage out of the high-risk region through architecture and recorder changes.

Spaceflight & Mission Analysis — Ansys STK/Astrogator, MAE 341 **Fall 2025**

- Built an Artemis-class cislunar trajectory in Astrogator and a single-station downlink study: identified 18 Kennedy Space Center contact windows totalling 166.65 h (42.8% of scenario), ~440,000 km worst-case slant range and ~19.75 dB worst-case E_b/N_0, showing the modelled link was access-limited rather than power-limited.
- Validated the orbital analysis against 3 nights of ISS observations, estimating a 93.334 ± 0.088 min period within ~0.5% of the 92.903 min mean-motion reference, and identified which of 3 reduction methods fails and why (repeat-pass timing ran 5.6% high on viewing geometry and horizon threshold).

Automated Railway Control System — MAE 412 **Spring 2025**

- Built a hands-free unload/sort/reload platform using 3 Hall-effect sensors, 2 rail switches, 2 servos, 1 relay and 3D-printed mechanisms to handle 4 stainless-steel balls across 2 sizes, sequenced in assembly over an Arduino ↔ ACIA serial link; achieved all objectives plus the stretch reloading goal with repeatable class-demonstration operation.

HONOURS, FELLOWSHIPS & FUNDING

John Marshall II Memorial Senior Thesis Fund — 2026 | Competitive proposal funding for OSCAR MARSKIN.
ACEE Summer Internship Program / Peter B. Lewis Fund — Summer 2025 | Competitive funded energy and environment research internship.
Summer Social Impact Internship (SSII) Fund — Recipient, 2×
Academic Excellence (Rwanda) — First Lady Academic Excellence Award, 2018 & 2022; Rwanda Education Board Award, 2021 (2nd nationwide, A-Level, Gashora Girls Academy of Science & Technology); AIMS Outstanding Achievement Award, 2019 (4th nationally, Mathematics).
Languages — English, French, Kinyarwanda, Kirundi, Kiswahili, Spanish, German